

Anisotropic cellulose nanocrystal hydrogel with multi-stimuli response to temperature and mechanical stress

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ABSTRACT

Conventional hydrogels with isotropic polymer networks usually lack selective response to external stimuli and that limits their applications in intelligent devices. Herein, hydrogels with distinctive anisotropic optical characteristics combined with thermosensitivity were prepared through in situ photopolymerization. Self-assembled cellulose nanocrystals (CNCs) with chiral nematic ordered structure were embedded in polyethylene glycol derivatives/polyacrylamide polymer networks. The arrangement of CNCs showed a strong dependence on the self-assembly angle and standing time, enabling the fabrication of hydrogels with customizable CNCs arrangements. Increasing the self-assembly angle from 0° to 90° changed the CNCs arrangement from chiral nematic to symmetrical nematic order which, together with CNCs dynamic arrangement from isotropic to annealed chiral nematic phase at longer standing time, provided versatile ways to produce CNCs hydrogels with tunable anisotropic properties. In addition, the obtained hydrogel displayed reversible temperature and compression response, showing excellent promise to be used as soft mechanical stress and temperature sensors or novel anti-counterfeiting materials.

1. Introduction

Hydrogel, a soft and hydrophilic material with three-dimensional network structure, has been widely used in environmental protection, medicine, agriculture, and architecture. Recently, hydrogels with stimuli-responsive properties are being intensively studied for versatile applications as intelligent devices. However, polymer network obtained from the conventional polymerization process shows uniform structure in all directions, resulting in an isotropic response to external stimuli. Therefore, it is highly desirable to develop hydrogels with oriented structures and anisotropic properties. One of the typical strategies for fabricating oriented structures in amorphous polymers is to introduce nanoparticles coupled with induced alignment under mechanical force (Mredha et al., 2018; Ye et al., 2017), magnetic field (Yook et al., 2019), electric field (Yang et al., 2018; Zhu et al., 2020), or other conditions (Lin et al., 2016; Wu et al., 2017). The resulting highly ordered hydrogels are expected to possess distinctive anisotropic optical, mechanical,

electrical, or other functional performance, suitable for applications in intelligent devices, such as actuators (Le et al., 2019; Sun et al., 2018), sensors (Hiratani et al., 2018), optical devices (Qiao et al., 2019), and artificial muscles (Park & Kim, 2020).

Although the above strategies could spatially orient nanoparticles in a polymer matrix, the orientation process relied on external forces, which could become impractical for constructing anisotropic materials of large sizes. Inspired by the natural formation of ordered structures, such as protein folding and double helix structure of nucleic acid, the self-assembly technique has been adopted as a bottom-up strategy to assemble disordered particles into ordered states for constructing new functional materials (Lan & Wang, 2016; Santiago-Gonzalez et al., 2018). During the self-assembly process, particles ranging from micro- and mesoscale to macroscale aggregate spontaneously to form thermodynamically stable blocks without the need for human intervention (Whitesides & Grzybowski, 2002). Qin et al. (2019) fabricated a nanocomposite hydrogel consisting of a highly ordered self-assembled

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lamellar structure of precious metal nanoparticles through in situ polymerization. The well-oriented nanostructures imparted highly anisotropic mechanical properties to the hydrogel with the tangent modulus parallel to the oriented lamellae 2.9 times higher than the tangent modulus in the vertical direction. [Pei et al. \(2015\)](#) employed the spatial confinement method for inducing self-assembly of the rod-like virus to form the cholesteric phase in a polymeric hydrogel precursor before the gelation process. Even if the virus was etched away, the imprinted chiral structure formed by the virus endowed the hydrogel with three-dimensional (3D) chiral ordered cylindrical pores, showing anisotropic optical properties with stimuli responsiveness.

In order to reduce our reliance on fossil fuel, materials derived from renewable resources are increasingly explored as substitutes with a higher sustainability ([Tardy et al., 2021](#)). In particular, cellulose-based nanomaterials are highly promising owing to their remarkable flexibility, biodegradability, optical transparency, and high mechanical strength. Furthermore, cellulose nanocrystals (CNCs) have the ability to undergo spontaneous self-assembly, which has provided new opportunities to construct hierarchical and functional macroscopic structures with highly attractive properties ([Ajadary et al., 2020](#)). The cholesteric phase is the most common self-assembly structure spontaneously formed by CNCs in suspension. This anisotropic phase consists of stacked layers of CNCs. In each layer, CNC rods are in parallel alignment along a vector. The vector directions of adjacent layers rotate at a certain angle around the perpendicular axis and form a helix ([Beck et al., 2013](#); [Parker et al., 2018](#)). The pitch is commonly defined as twice the perpendicular distance between two nearby layers with the same vector orientation. Interestingly, CNC assemblies show visible iridescent color for a sufficiently small pitch; otherwise, only interference color can be observed under crossed polarizers. [Charlotte et al. \(Boott et al., 2020\)](#) showed the reversible transformation of chiral nematic CNC elastomer from colorless to colored due to the reduction of the helical pitch of CNCs under the mechanical stretching or compression process.

Although self-assembled CNCs have attractive birefringence performance, most applications are limited to preparing CNC films and chiral nematic templates. Therefore, to explore the application of aligned CNCs, many previous studies have introduced polymers into CNC ordered structures; thus, endowing CNC composite with flexibility, processability, and durability. [Kose, Tran, Lewis, Hamad, & MacLachlan \(2019\)](#) prepared a CNC elastomer with a reversible optical effect by embedding CNC in a copolymer of ethyl acrylate and 2-hydroxyethyl acrylate. When stretched, the chiral nematic structure of the CNC elastomer changed into a pseudonematic arrangement, along with dramatic interference color changes. They also obtained shear-aligned pseudonematic CNCs in a poly (ethyl acrylate) elastomer ([Kose, Boott, Hamad, & MacLachlan, 2019](#)). Different stretching directions caused discrepancies in CNC alignment due to the long-range anisotropy of CNCs, resulting in anisotropy birefringence.

However, the self-assembly of CNC in polymer precursor suspension has been seldomly reported. A typical approach for introducing CNC chiral nematic structure into composite materials is to swell or infiltrate CNC films with monomer or polymer solutions. But the infiltration process has high requirements on the flowability, concentration, and molecular weight ([Bast et al., 2021](#)). In addition, the strong interactions between CNCs caused by the hornification effect will reduce the accessibility of CNCs, which may lead to inadequate interfacial interactions between the CNC and the molecular chains of the infiltrating polymer ([Qijun et al., 2018](#)). Therefore, in this study, an anisotropic multi-stimuli hydrogel was prepared through in situ photopolymerization of two oligo (ethylene glycol) methacrylates with different side-chain lengths and a small amount of acrylamide (AAM). Self-assembled CNCs with helical chiral nematic structures were embedded in thermosensitive polymers. The obtained hydrogels displayed anisotropic optical performance and thermosensitivity. The crosslinking density was changed by altering the content of AAM and polymers/CNC ratio, respectively. The corresponding chemical structure, water state, viscoelasticity, and

micromorphology were characterized and discussed. The self-assembly of CNCs occurred directly in the polymer precursor. Additionally, the self-assembly process could be investigated from a longitudinal and cross-sectional view of the sample. Furthermore, the mechanical response property was studied, and the possible temperature and mechanical sensing mechanism were discussed.

2. Material and methods

2.1. Materials

Di (ethylene glycol) methyl ether methacrylate (DEGMA, Mw = 188.22), poly (ethylene glycol) methyl ether methacrylate (PEGMA, Mn = 500), 2-Hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (photoinitiator), and N, N'-Methylenebisacrylamide (crosslinker) were purchased from Sigma-Aldrich. Cellulose nanocrystal (CNC, Mw = 14,700–27,850) suspension was purchased from CelluForce. Acrylamide (AAM) was purchased from Caledon Laboratories. Deionized water was used. All reagents were used without further purification.

2.2. Preparations of hydrogels

Hydrogels were prepared by mixing 5 wt% CNC suspension, monomers, photoinitiator, and crosslinker in vials using a magnetic stir for 1 h. The obtained homogeneous mixture was then injected into customized sealed cells (made out of two pieces of glass slides with EVA foam sheet as the spacer) and allowed to stand for 48 h (standing time). The photo-polymerization was conducted by using an ultraviolet (UV) light source (10 W, 365 nm) for 2 h. The contents of CNC/PEGMA/DEGMA/AAM in hydrogels are shown in [Table 1](#). The amount of the photoinitiator was 0.5% (w/w) of the total monomers (PEGMA, DEGMA, and AAM), and the crosslinker was 10% (w/w) of AAM. To remove the unreacted monomers, prepared hydrogels were soaked in water for 3

Table 1
The formula of samples with different components.

No.	Sample name	5 wt % CNC (or H ₂ O)	PEGMA: DEGMA: AAM (g: g: g)	Photoinitiator (mg)	Crosslinker (mg)
I	C ₁ P _{1.17} D _{2.33}	20 g CNC	1.17: 2.33: 0	17.5	–
II	C ₁ P ₁ D ₂ A _{0.5}	20 g CNC	1: 2: 0.5	17.5	50
III	C ₁ P _{0.875} D _{1.75} A _{0.875}	20 g CNC	0.875: 1.75: 0.875	17.5	87.5
IV	C ₁ P _{0.78} D _{1.56} A _{1.17}	20 g CNC	0.78: 1.56: 1.17	17.5	117
V	C ₁ A _{3.5}	20 g CNC	0: 0: 3.5	17.5	350
VI	P ₁ D ₂ A _{0.5}	20 g H ₂ O	1: 2: 0.5	17.5	50
VII	PAAM	20 g H ₂ O	0: 0: 3.5	17.5	350
VIII ^a	P _{1.16} D _{2.34}	20 g H ₂ O	1.16: 2.34: 0	17.5	–
IX	C ₁ P ₂ D ₄ A ₁	20 g CNC	2: 4: 1	35	100
X	C ₁ P ₃ D ₆ A _{1.5}	20 g CNC	3: 6: 1.5	52.52	150
XI ^b	PD	–	1.16: 2.34: 0	17.5	–

^a Sample VIII was in liquid state, so it was not tested by Fourier Transform Infrared (FTIR) Spectroscopy.

^b Sample XI was gel-like and was only used for FTIR analysis as a reference to determine if there was any chemical bond between PEGMA-co-DEGMA chains and other components in CPDA hydrogels.

days while changing the water twice a day. Samples were named as CxPxDxAx, where the subscript x was the mass of each component (Table 1). CPDA was the abbreviation of hydrogels prepared by CNC, PEGMA, DEGMA, and AAm. PD represented the hydrogel prepared by PEGMA and DEGMA.

2.3. Characterization

The polarized optical appearance of the hydrogels was investigated using two crossed polarizers on a CKX41 Inverted Microscope (Olympus, Japan). The microscopic morphology was recorded with a Field Emission Scanning Electron Microscope (FE-SEM, ZEISS Sigma 300, Germany) with the sample gold sputter-coated. The chemical structure was evaluated by Spectrum Two FTIR Spectrometer (PerkinElmer, USA). The dynamic mechanical property was measured with a Dynamic Mechanical Analyzer Q800 (DMA, TA Instruments, USA) in the multi-frequency-strain mode. Swollen hydrogel specimens were cut into cuboids with a size of around 12 mm × 4.5 mm × 2 mm. The DMA test was conducted with the temperature increasing from 30 °C to 80 °C at a rate of 5 °C/min. Thermal analysis was performed by Differential Scanning Calorimetry Q2000 (DSC, TA Instruments, USA). Swollen hydrogels were tested in high-pressure capsules from −50 °C to 60 °C, at a heating rate of 10 °C/min. The freezing water content in total water was calculated as the melting enthalpy ratio of the water in the sample (ΔH_f) and pure water (ΔH_p). The melting enthalpy was integrated from the endothermic peak area by the TA Universal Analysis software.

$$C_f(\%) = \frac{\Delta H_f}{\Delta H_p} \times 100 \quad (1)$$

$$C_b(\%) = (1 - C_f) \times 100 \quad (2)$$

where C_f and C_b are freezing and bound water contents in total water, respectively. The freezing and bound water contents in the hydrogel

were calculated as following:

$$C_{fh}(\%) = C_f \cdot \omega \cdot 100 \quad (3)$$

$$C_{bh}(\%) = C_b \cdot \omega \cdot 100 \quad (4)$$

$$C_{dh}(\%) = (1 - C_{fh} - C_{bh}) \times 100 \quad (5)$$

where C_{fh} and C_{bh} represent freezing and bound water contents in the hydrogel, respectively. ω is the total water content of hydrogel. C_{dh} is the content of dry gel in the hydrogel.

The thermosensitive properties were evaluated according to the equilibrium swelling ratio (ESR) at different temperatures (30 °C, 40 °C, 50 °C, 60 °C, 70 °C, and 80 °C). The ESR was calculated as following Eq. (6):

$$ESR(\%) = \frac{W_e - W_0}{W_0} \times 100 \quad (6)$$

where W_e is the mass of the hydrogel reached the swelling equilibrium at a specific temperature, and W_0 is the mass of the dry hydrogel.

3. Results and discussion

Rod-like CNCs can form hierarchical self-assembly structures in suspensions spontaneously under certain conditions (Bai et al., 2020). However, the continuous phase of CNC assembly is easily affected by external fields in the fluid. In this study, CNCs were fixed in the composite hydrogels through in situ photopolymerization. Fig. 1a showed that 5 wt% CNC water suspension and monomers with initiator mixed well in the liquid state. After 2 h of UV-radiation, samples were cured, except for I and VIII. Comparing the composition of the cured samples, neither I nor VIII had polyacrylamide, indicating that polyacrylamide (PAAm) played an important crosslinking role during the hydrogel polymerization process.

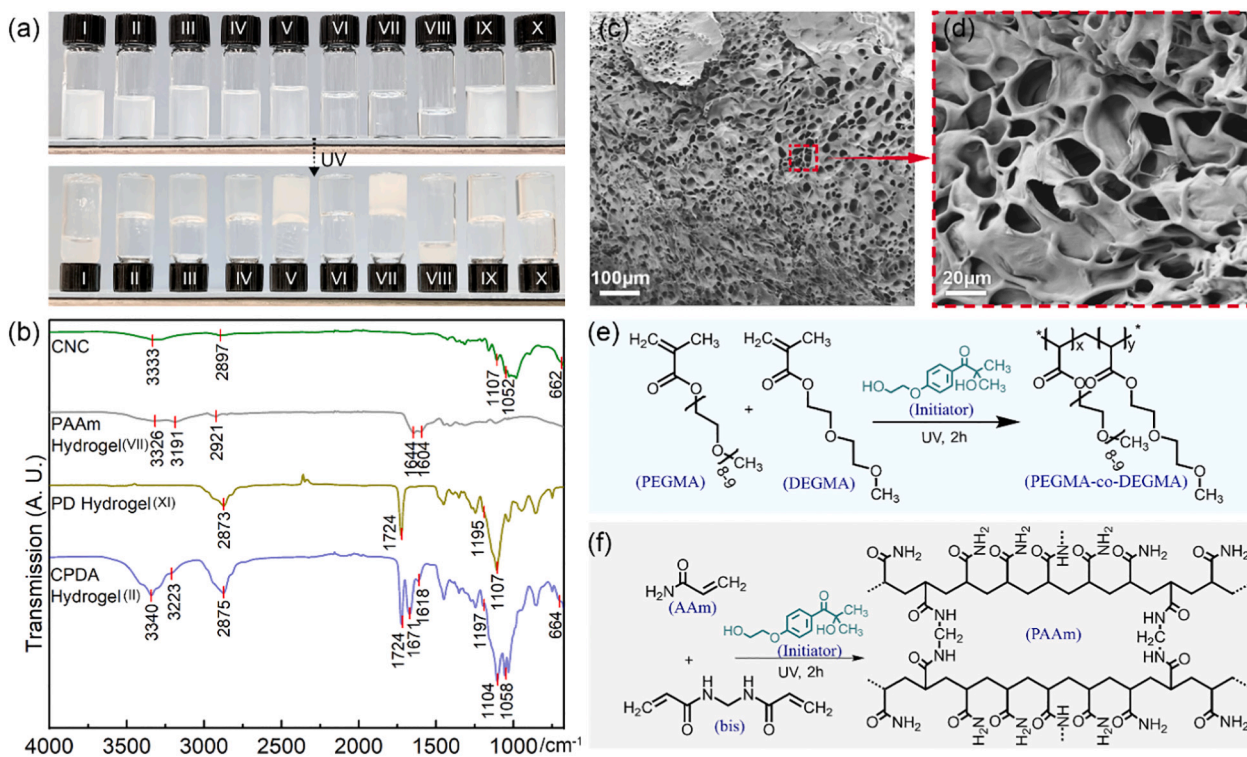


Fig. 1. (a) Photograph of the pre-reaction mixture (top) and gel compounds (bottom) formed before and after UV triggered photo-polymerization. (b) ATR-FTIR spectra of CNC film, PAAm hydrogel (VII), PD hydrogel (XI), and CPDA hydrogel (II). (c) and (d) are SEM images of fractured cross-sectional surface of CPDA hydrogel (sample II). (e) and (f) are proposed synthesis reactions during the hydrogel preparation.

To elucidate the chemical structures of the CPDA composite hydrogels, FTIR analysis was conducted. As shown in Fig. 1b, the CNC spectrum had peaks at 3333, 662, 1052, and 1107 cm^{-1} for O—H stretching, O—H bending, C—O stretching, and C—O—C stretching, respectively. In the spectrum of PAAm hydrogel (sample VII), peaks at 3326 and 3191 cm^{-1} were absorption peaks of N—H stretching, and the peak at 1604 cm^{-1} was the in-plane N—H bending vibration. The peak at 1644 cm^{-1} was the stretching vibration of C=O in the amide group. In the spectrum of PD hydrogel (sample XI), peaks at 1724 and 1195 cm^{-1} could be separately attributed to the C=O and C—O stretching vibration of the ester group; the peak at 1107 cm^{-1} was attributed to C—O—C stretching vibration. Comparing the above three spectra, no new characteristic peak appeared in the CPDA hydrogel (sample II); however, peaks of N—H stretching (3340 cm^{-1}) and C=O stretching (1671 cm^{-1}) shifted to higher wavenumbers. This blueshift might be caused by the lower density of intermolecular hydrogen bonding in the composite hydrogel. Furthermore, the peak of O—H stretching was not obvious, possibly covered by the N—H stretching peak.

The microscopic morphology showed that CPDA composite hydrogel had a network structure (Fig. 1c, d). This network consisted of PEGMA-co-DEGMA and PAAm with CNC particles embedded (Liu et al., 2016). Fig. 1e showed that PEGMA-co-DEGMA was a linear polymer with long side chains. The linear polymer chains could form a 3D framework through the self-entanglement effect. But, in samples I and VIII, PEGMA-co-DEGMA was in low density that could not support its linear structure forming a 3D solid gel. Hence, the introduction of the PAAm network played an essential role in connecting linear chains of PEGMA-co-DEGMA and integrated them into a 3D network structure (Fig. 1f).

To further investigate the influence of PAAm on CPDA hydrogels, several hydrogels were prepared with an increased percentage of AAm content. Under the premise of fixing the total amount of monomers (including AAm, PEGMA, and DEGMA) as 3.5 g, AAm contents were 0.5, 0.875, 1.17, and 3.5 g in samples II, III, IV, and V, respectively. As the AAm content increased, one of the N—H stretching peaks shifted from 3223 to 3192 cm^{-1} , and the peak of the C=O stretching shifted from

1671 to 1645 cm^{-1} (Fig. 2a). These redshifts were due to the higher content of hydrogen bonding from the increased amide groups. The viscoelasticity was obtained from the DMA testing results. Fig. 2b showed that the storage modulus (G') was higher than the loss modulus (G''), indicating that the obtained hydrogel was an elastomer. G' and G'' at a set temperature increased with a higher content of AAm, and no obvious peak appeared in the curves of $\tan\delta$. The higher the G' , the greater was the elastic stiffness, while the higher the G'' , the larger the viscoelastic response (Zeltmann et al., 2017). According to the ESR results in Fig. 2c, the initial ESR data (30 °C) of samples were in descending order as the AAm content increased. A lower ESR indicated a higher crosslinking density (Omidian et al., 1994; Zhou et al., 2014). Therefore, the stiffness reinforcement and the viscoelastic enhancement were mostly attributed to the increased crosslinking density and hydrogen bonding.

Moreover, G' and G'' positively correlated with temperature. The temperature dependence could be associated with the thermal sensitivity of the sample. In Fig. 2c, samples II, III, and IV showed thermo-sensitive properties with their ESR decreasing with the increase in temperature. According to the literature, ESR was related to the conformation of polymer chains (Liu et al., 2016). Hydrophobic groups on polymer chains were driven by the entropy increment to assemble when the temperature increases. Additionally, increasing the molecular thermal motion made hydrogen bonds between hydrophilic groups and water molecules easier to be broken. Then, water could be expelled (Fig. 2d), ESR decreased, and the volume of the hydrogel shrunk. The denser polymer chains in samples II, III, and IV were responsible for the higher G' and G'' . However, the ESR curve of sample V almost remained flat with varying temperature, indicating that sample V had almost no temperature sensitivity. In contrast, its G' and G'' showed the strongest temperature dependence. Thus, the thermosensitive property was not the major factor that affected the complex modulus of the hydrogels. In terms of polymer viscoelasticity, with the increase in temperature, intermolecular hydrogen bonds broke and molecular segmental motion intensified. Molecular chains had tendencies to slip under stress;

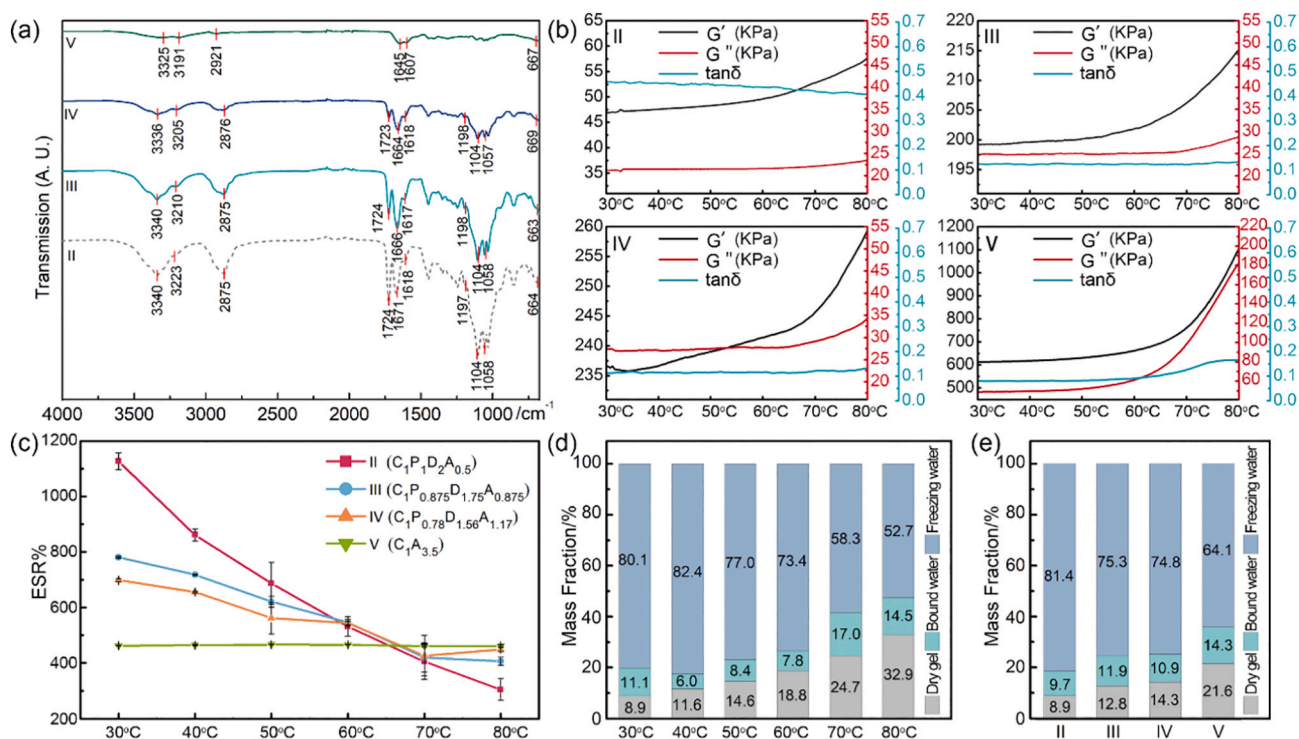


Fig. 2. ATR-FTIR spectra (a), dynamic mechanical analysis (b), and thermosensitive properties (c) of CPDA hydrogel II, III, IV, and V. (d) Water state and content in CPDA hydrogel II at different temperatures. (e) Water state and content in CPDA hydrogels II, III, IV, and V at room temperature (25 °C).

however, crosslinking points prevented these slippages, generating more heat with a higher G'' value.

The state of water can directly reflect the behavior of hydrogels, and it is necessary to illustrate the interactions between the polymer network and water. Generally, water has three states in hydrogels: bound, intermediate, and free water. Bound water is nonfreezing water, which is tightly bound to polymers through hydrogen bonding. Free and intermediate water are freezing water. Free water does not participate in hydrogen bonds with the hydrogel polymer, behaving as pure water, while intermediate water has a weak interaction with polymers. Thus, the intermediate water shows a lower phase transition temperature than free water.

Furthermore, water contents have been calculated to obtain the information of water states in hydrogels according to the melting enthalpy. Fig. 2d showed that, as the gel content increased from 8.9 wt% to 32.9 wt% along with the increasing temperature, the total water content (freezing and bound water) decreased. At higher temperatures,

the content ratios between the freezing water and dry gel content were lower (Supplementary Table S1). This can be explained by the shrinkage of the hydrogel polymer network by expelling the excess water. Meanwhile, as the temperature increased from the initial temperature to 40 °C, the ratio of bound water and dry gel content decreased from 1.25 to 0.52 and remained around 0.4–0.7 with an increasing temperature (Supplementary Table S1). This change was related to the structural conversion during the temperature response stage. It is well-known that hydrogels form dense skin to prevent water loss at the initial temperature rise. In this period, the chemical potential accumulates and generates strong pressure to push water outside. Then, a part of the bound water converts into the semi-bound state under the pressure-induced shear effect (Pasqui et al., 2012). After that, the pressure becomes steady. As shown in Fig. 2c, sample II showed an approximately linear ESR-temperature relationship from 40 °C to 80 °C. The degree of crosslinking significantly affected the water state in hydrogels. The ratio of freezing and bound water content to dry gel content decreased in the

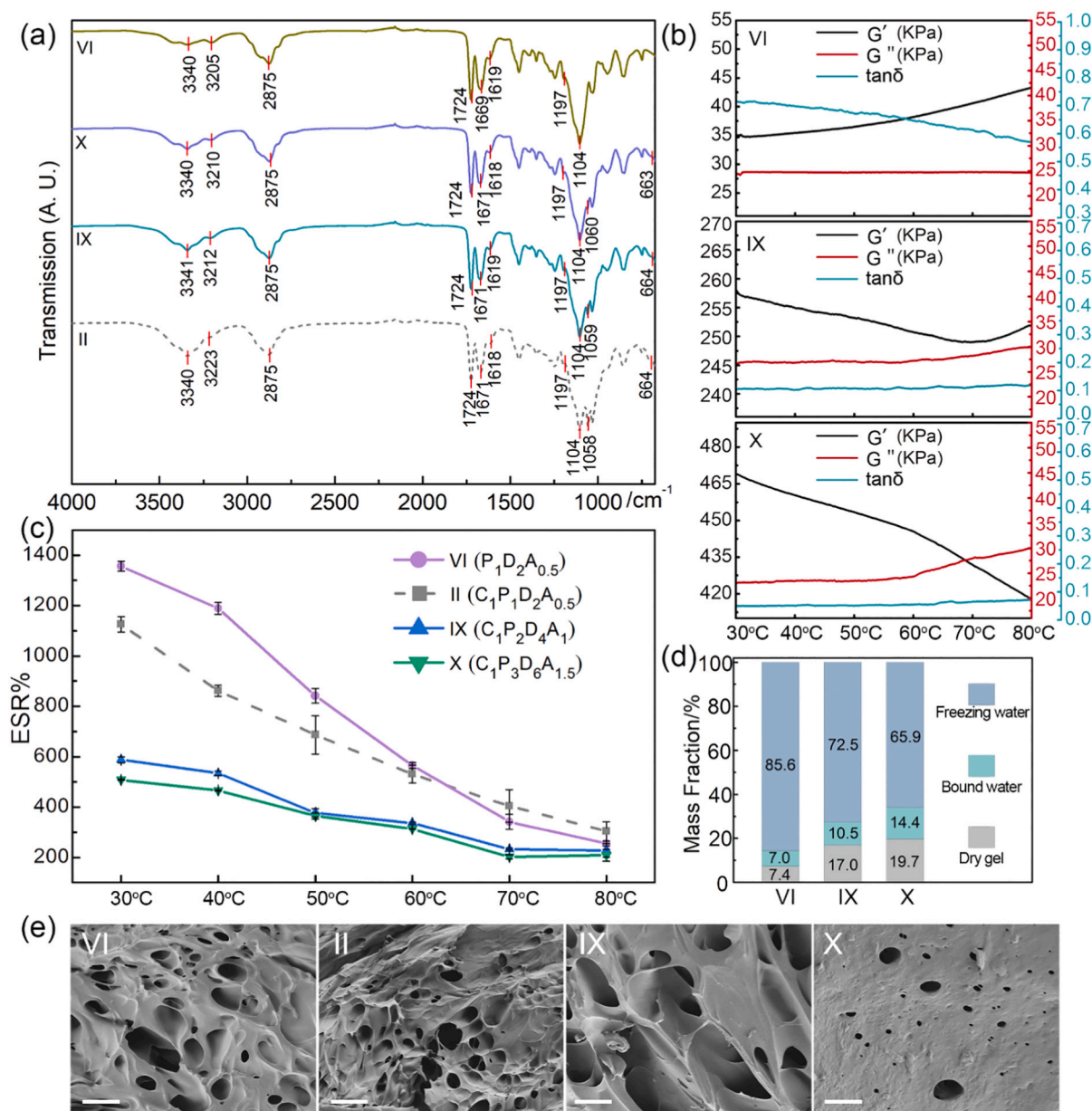


Fig. 3. ATR-FTIR spectra (a), dynamic mechanical analysis (b), thermosensitive properties (c) of CPDA hydrogel VI, IX, and X. (II for reference) (d) Water state and content in CPDA hydrogels VI, IX, and X at room temperature (25 °C). (e) SEM images of sample VI, II, IX, and X (scale bar is 50 μm). CNC contents in samples II, IX, X, and VI are 22.2, 12.5, 8.7, and 0 wt%, respectively (Supplementary Table S2).

order of II > III > IV > V (Fig. 2e). Although AAm polymer is hydrophilic, a higher AAm content in the precursor yielded a lower water content in the hydrogel. The reason was that an increasing AAm content led to a higher crosslinking density in the hydrogel, reducing the water absorption.

The polymeric chains induced confinement on CNC particles due to the large amount of CNCs dispersed in the hydrogel polymer network. Thus, the crosslink density can be treated as the indication of the confinement degree for CNCs. In addition to changing the AAm content, altering the mass ratio between CNCs and polymers was another way of adjusting the confinement degree for CNCs. Hydrogel samples with varied CNC contents showed similar FTIR curves in Fig. 3a, except for sample VI, which lacked two characteristic peaks associated with C—O stretching for CNC at around 1052 cm^{-1} and O—H bending at 662 cm^{-1} . The peak of the N—H stretching shifted from 3223 to 3205 cm^{-1} due to the lack of CNC to create obstacles or hindrances between the two polar functional groups. Therefore, the number of hydrogen bonding between —NH_2 groups increased, resulting in the redshift of N—H stretching.

Compared with samples containing CNCs, sample VI (without CNCs) had the largest ESR at the initial temperature (Fig. 3c). However, ESR of samples II, IX, and X decreased as the CNC content decreased, showing the same trend in freezing and bound water content in Fig. 3d. This is because of the formation of physical bonds between CNCs and polymer chains; thus, increasing the crosslinking density and decreasing ESR. A higher polymer content increased the chemical crosslinking density. Fig. 3e showed that, as the polymer content increased from 77.8 (sample II) to 91.3 wt% (sample X), the hydrogel became denser, showing less pore (Polymer contents were calculated and given in the Supplementary Table S2).

In Figs. 2b and 3b, the initial storage modulus of samples VI, II, IX, and X increased sequentially, showing an opposite trend to ESR in Fig. 3c. These results indicated that the higher the crosslinking density,

the greater was the stiffness of the hydrogel. G'' of samples II, IX, and X exhibited temperature dependence, increasing as the temperature increased, whereas G'' of sample VI didn't show fluctuation with the increasing temperature. This demonstrated that there were physical bonds, such as hydrogen bonds, between CNCs and polymer molecules. For samples II, IX, and X, which contained CNCs, the hydrogen bonds between CNCs and polymer chains were broken when the temperature was high, intensifying the segmental motion to generate more heat for a higher G'' . In contrast, sample VI (without CNCs) almost had no hydrogen bonds that few intermolecular hydrogen bonds were broken at the high temperature.

Anisotropic materials usually split polarized light into ordinary and extraordinary rays, which may not have the same velocity or pathway when traveling through the materials. As the two rays emerge from an anisotropic material, interference colors are produced. Birefringence can be used to describe the difference in velocity of the two rays (Gray, 2016). Low birefringent materials exhibit only white, gray, and black interference colors. As the birefringence increases, higher-order interference colors appear (Haywood & Davis, 2017). A higher degree of alignment will increase the system's birefringence for a nanorod system, such as CNC suspension. If the alignment is uniform, the interference color will also be uniform. The interference color and local orientation of CNC rods in suspensions or films can be observed using a cross-polarized optical microscope (Gray, 2016). However, due to surface anchoring and boundary effect, introducing a polymer network might affect the birefringence or texture of a CNC self-assembly system by changing the orientational ordering parameters of CNCs (Yang et al., 2013).

To explore the orientation and alignment of CNCs in the polymer network, hydrogel films prepared with different self-assembly angles ($\theta = 0^\circ, 45^\circ, 90^\circ$) were investigated through an optical microscope between crossed polarizers (see Fig. 4). Samples for polarized light micrographs were taken from the center of the hydrogel films to avoid

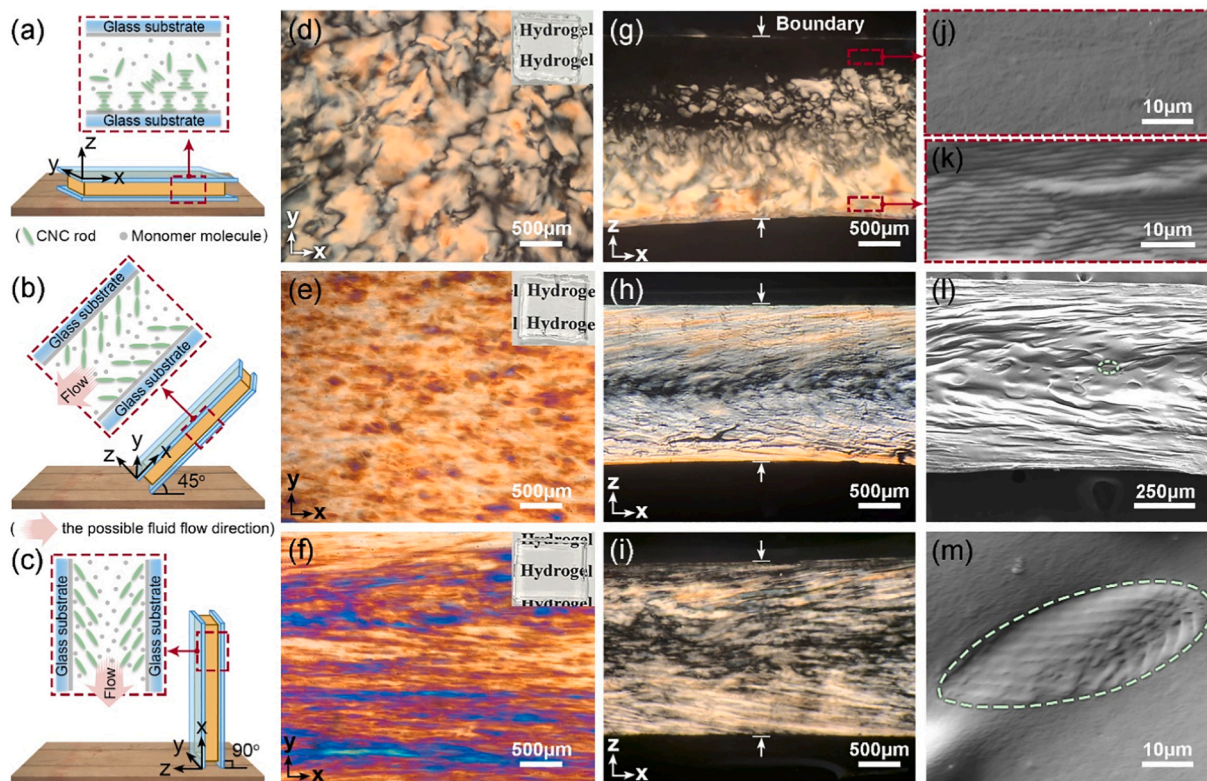


Fig. 4. Schematic of CNCs standing for 48 h at self-assembly angles of 0° (a), 45° (b), 90° (c), and the corresponding cross-polarized optical micrographs viewed from x-y plane (d, e, f; the inserted images are photos of hydrogels under visible light) and x-z plane (g, h, i). (j) and (k) are SEM images of the isotropic (upper) and anisotropic (lower) area of (g), respectively. (l) is the SEM image of (i), and (m) is the enlarged view of the green ellipse marked in (l). The white arrows in (g, h, i) indicate the upper and lower boundaries of the samples.

boundary effects. Contrary to CNC assemblies exhibiting visible iridescent color, hydrogels prepared in this study were transparent, having excellent light transmission rate under natural light (Fig. 4d–f and supplementary Fig. S1). They showed typical planar (Fig. 4d) and fingerprint texture of cholesteric phase (Supplementary Fig. S2) in polarized light, indicating that CNC rods formed chiral nematic helix in the hydrogel. The texture difference shown on the x-y and x-z planes was relevant to the optical axis direction of CNC assembly and microscope. Bright areas in Fig. 4d were birefringent regions, and dark lines between adjacent bright domains were disclination caused by discontinuous CNC orientation (Saha & Davis, 2018). The phase separation was clearly observed from the x-z plane (Fig. 4g) with the presence of a dark isotropic region at the top and a bright anisotropic region at the bottom. The isotropic phase that exhibited flat micromorphology was likely the polymer matrix (Fig. 4j), while the anisotropic phase with classical annealed chiral-nematic structure was the stacked CNCs layers (Fig. 4k).

The phase separation of CNCs suspension occurred above a critical concentration. As the phase separation took place, CNCs aggregated to form a short-range ordered structure (tactoid). Given sufficient standing time, the system reached a dynamic equilibration during which the tactoids fused and precipitated to form a long-range ordered anisotropic phase (Klockars et al., 2018).

When increasing θ to 45° or 90° , a symmetrical nematic phase was observed in the samples showing oriented textures from the x-y (Fig. 4e, f) and x-z planes (Fig. 4h, i) using the centerline as the axis. Strips near the boundary (Fig. 4l) demonstrated a preferred alignment along the substrate's surface, indicating that CNC rods arranged in quasi-parallel orientation to the axis x. The dark area was larger at the center of the sample than at the edge. Some tactoids (i.e. Fig. 4m) had been observed in the central domain (Nyström et al., 2018). Hence, the results obtained here suggested that the CNCs content at the center was lower than that at the edge, thereby forming only short-range ordered tactoids but not

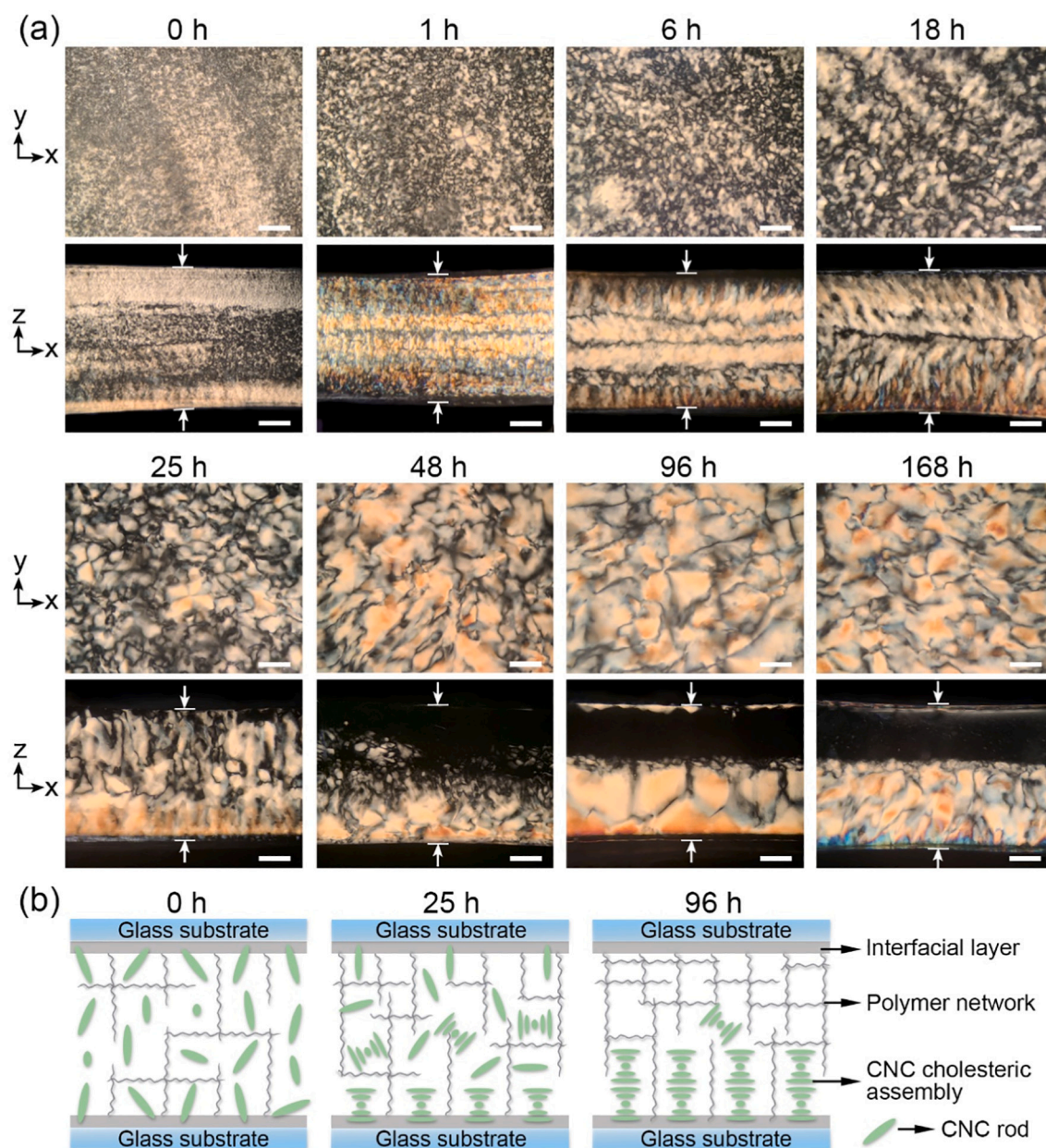


Fig. 5. (a) Cross-polarized optical microscope images of CPDA hydrogels prepared after standing for 0, 1, 6, 18, 25, 48, 96, and 168 h, respectively. (b) Schematic of proposed CNC arrangement in the polymer network after UV curing with standing times respectively of 0, 25 and 96 h. All scale bars are 500 μm , and white arrows indicate the upper or lower boundaries of the samples.

continuous and ordered phases through self-assembly. We suggest that the segregation of CNCs was caused by interactions between glass and CNCs and that produced an anchoring effect. Due to the spatial decay oscillations of the anchoring force, the farther away from the substrate surface, the weaker the anchoring force (Jerome, 1991). Thus, the fluid state precursor flowed easily in the middle layer as a result of the gravity, especially when the cell was orientated. As shown in Fig. 4b and c, one end of the CNC rod appeared as anchored to the substrate while the other end pointed to the fluid flow direction, forming such a unique discontinuous nematic phase (Chen et al., 1996; Chen & Gray, 2002; Geng et al., 2018; Wang & MacLachlan, 2018).

The CNC self-assembly process in a monomer suspension was similar to that in water. Fig. 5a illustrated the evolution of CNC arrangement in the monomer precursor suspension during the self-assembly process. The images consisted of bright and dark dots at 0 h, suggesting that the sample had a low optical path difference. It meant CNC assemblies had low or non-helix structure at 0 h (Kose, Tran, et al., 2019). As the standing time increased, the CNC rods began to gather and formed bright domains with vivid interference color, as shown in the x-y plane in Fig. 5a. Later, phase separation occurred after 48 h of equilibration, forming an isotropic/anisotropic biphasic state in the x-z plane. However, owing to the boundary effect, the CNC alignment in the confinement space between two plates differed from that in bulk. From the x-z plane, images of 0, 1, and 6 h, had a similar symmetrical banding texture with dark linear defects between bands. These linear defects gradually reduced from 6 to 18 h, except for the one at the horizontal symmetric axis. Additionally, horizontal linear defects were almost invisible at around 25 h. It demonstrated that this CNC/monomer suspension was unstable. With the descend of CNC rods under gravity and the rotation under surface charges, the bands were gradually merged; thus, the linear defects as the boundary lines blurred gradually and then disappeared. Fig. 5b displayed the possible arrangement of CNC after various standing times. When the cell with precursor suspension was placed under UV light, the polymerization completed first at the interfacial layer. Initially (at 0 h), CNCs were randomly dispersed in the interfacial layers, showing direction-independent birefringence on the x-y plane. As the equilibration time increased, CNCs aggregated into tactoids, which descended and merged gradually to form a local anisotropic phase at the bottom (25 h). With a longer equilibration time, the self-assembly process of CNCs reached an equilibrium in the suspension, presenting a biphasic state with an upper isotropic and a lower chiral nematic phase as shown in the cured hydrogel (96 h).

There was a strong relationship between the interference color and CNC rod orientation (Kose, Tran, et al., 2019). Since the hydrogel was compressed on the x- or y-axis, it displayed vivid interference color changes on the z-axis in Fig. 6. According to the above analysis, CNC in the sample (II) was assumed to have a planar chiral nematic structure with the rods parallel to the x-y plane. Blue, green, yellow, and red colors arose sequentially as the compression proceeded. The color transformation followed the same sequence as the second-order color of the Michel-Lévy color chart when the birefringence increased. This outcome suggested that compression increased the birefringence, demonstrating that CNCs experienced a higher degree of alignment during the compression process (Haywood & Davis, 2017; Kose, Boott, et al., 2019; Kose, Tran, et al., 2019). This phenomenon was reversible such that the interference color and hydrogel shape returned almost to the original once the external force was removed. The average red-green-blue (RGB) value of each image was obtained from Photoshop. The average color was shown in the image according to the International Commission on Illumination (CIE) RGB color space (Fig. 6b and Supplementary Fig. S3).

In addition to mechanical compression, altering the water content caused the deformation in the hydrogel, which led to changes in the orientation of CNCs. As shown in Fig. S4, the hydrogel significantly reduced in size as temperature increased from 30 °C to 80 °C. However, the interference colors in the x-y plane did not follow similar patterns to the ones observed during mechanical compression. Localized blue, green, red, and other color spots were observed when temperature increased. Furthermore, some dark linear defects were observed but became smaller as the overall volume of the hydrogel decreased. The texture change during the heating process was similar to that in the reverse water absorption process (Supplementary Fig. S5). We suggest that isotropic shrinkage or swelling of the hydrogel preferentially occurred in the x-y plane due to the volume change caused by the alteration in water content which resulted in different interference patterns from what were seen under mechanical compression.

4. Conclusion

In summary, a facile approach for preparing anisotropic CNC containing hydrogels with multi-stimuli responses through in situ photopolymerization was developed in this study. The self-assembled CNCs were fixed in the thermosensitive polymer matrix and showed anisotropic optical behaviors under the crossed polarizers. The crosslinking

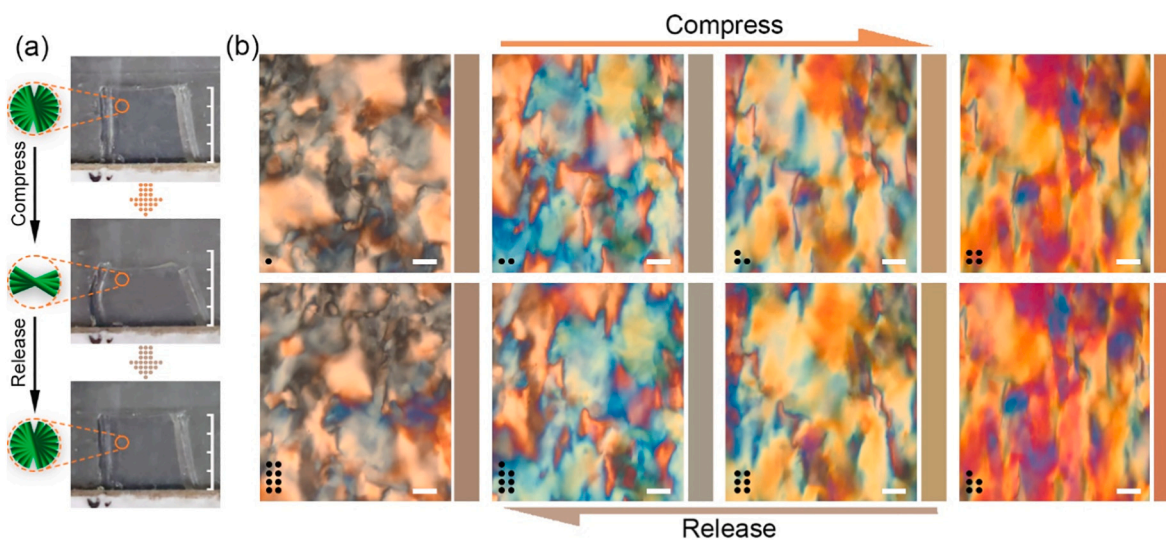


Fig. 6. Schematic of CPDA II hydrogel under the mechanical compress/release process (a) and the reversible mechanical response of interference color of the hydrogel between crossed polarizers (b). The right-side color bar in each image of (b) is the average color corresponding to the left-side image. Scale in (a) is 2 mm, and 250 μm in (b).

density was changed respectively by altering the AAm content from 0.5 g to 3.5 g and adjusting the ratio of polymers/CNCs with the CNCs content ranging from 22.2 wt% to 8.7 wt%. It was found that a higher content of AAm or a lower content of CNCs led to a higher crosslinking density in the hydrogel, causing the increase in G' and G'' . The self-assembly angle and standing time were shown to be effective factors that affected the ordered structure of CNCs. When the self-assembly angle increased from 0° to 90° , the arrangement of CNC rods changed from chiral nematic structure to symmetrical nematic structure. Owing to the thermosensitive polymer matrix, the formed hydrogel showed a response to temperature, and its volume and ESR decreased gradually with the temperature rising from 30°C to 80°C . Furthermore, mechanical compression of the hydrogels led to the increase in birefringence, and the interference color of the hydrogels showed a reversible mechanical response. The response to stress and temperature was independent of each other. These hydrogels demonstrate excellent potentials as mechanical stress and temperature multi-stimuli sensors. Moreover, the high transparency of the hydrogels under visible light makes them highly suited for anti-counterfeiting applications.

CRedit authorship contribution statement

All the authors listed contributed to the completion of this research. Liu Liu: Conceptualization, Methodology, Investigation, Writing - Original draft. Nicolas R. Tanguy: Investigation, Writing - Review and Editing. Ning Yan: Conceptualization, Methodology, Resources, Supervision, Funding acquisition, Writing - Review and Editing. Yiqiang Wu: Supervision, Funding acquisition. Xiubo Liu: Conceptualization, Funding acquisition. Yan Qing: Methodology, Visualization.

Declaration of competing interest

The authors declare no competing financial interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbpol.2021.119005>.

References

- Ajdary, R., Tardy, B. L., Mattos, B. D., Bai, L., & Rojas, O. J. (2020). Plant nanomaterials and inspiration from nature: Water interactions and hierarchically structured hydrogels. *Advanced Materials*, 33(28), 2001085.
- Bai, L., Huan, S., Zhao, B., Zhu, Y., & Rojas, O. J. (2020). All-aqueous liquid crystal nanocellulose emulsions with permeable interfacial assembly. *ACS Nano*, 14(10), 13380–13390.
- Bast, L. K., Klockars, K. W., Greca, L. G., Rojas, O. J., Tardy, B. L., & Bruns, N. (2021). Infiltration of proteins in cholesteric cellulose structures. *Biomacromolecules*, 22(5), 2067–2080.
- Beck, S., Bouchard, J., Chauve, G., & Berry, R. (2013). Controlled production of patterns in iridescent solid films of cellulose nanocrystals. *Cellulose*, 20(3), 1401–1411.
- Boott, C. E., Tran, A., Hamad, W. Y., & MacLachlan, M. J. (2020). Cellulose nanocrystal elastomers with reversible visible color. *Angewandte Chemie International Edition*, 59(1), 226–231.
- Chen, W., & Gray, D. G. (2002). Interfacial tension between isotropic and anisotropic phases of a suspension of rodlike particles. *Langmuir*, 18(3), 633–637.
- Chen, W.-L., Sato, T., & Teramoto, A. (1996). Measurement of the interfacial tension between coexisting isotropic and nematic phases of a lyotropic polymer liquid crystal. *Macromolecules*, 29(12), 4283–4286.
- Geng, Y., Jang, J. H., Noh, K. G., Noh, J., Lagerwall, J. P. F., & Park, S. Y. (2018). Through the spherical looking-glass: Asymmetry enables multicolored internal reflection in cholesteric liquid crystal shells. *Advanced Optical Materials*, 6(1), 1700923.
- Gray, D. G. (2016). Recent advances in chiral nematic structure and iridescent color of cellulose nanocrystal films. *Nanomaterials*, 6(11), 213.
- Haywood, A. D., & Davis, V. A. (2017). Effects of liquid crystalline and shear alignment on the optical properties of cellulose nanocrystal films. *Cellulose*, 24(2), 705–716.
- Hiratani, T., Kose, O., Hamad, W. Y., & MacLachlan, M. J. (2018). Stable and sensitive stimuli-responsive anisotropic hydrogels for sensing ionic strength and pressure. *Materials Horizons*, 5(6), 1076–1081.
- Jerome, B. (1991). Surface effects and anchoring in liquid crystals. *Reports on Progress in Physics*, 54(3), 391.
- Klockars, K. W., Tardy, B. L., Borghei, M., Tripathi, A., & Rojas, O. J. (2018). Effect of anisotropy of cellulose nanocrystal suspensions on stratification, domain structure formation, and structural colors. *Biomacromolecules*, 19(7).
- Kose, O., Boott, C. E., Hamad, W. Y., & MacLachlan, M. J. (2019). Stimuli-responsive anisotropic materials based on unidirectional organization of cellulose nanocrystals in an elastomer. *Macromolecules*, 52(14), 5317–5324.
- Kose, O., Tran, A., Lewis, L., Hamad, W. Y., & MacLachlan, M. J. (2019). Unwinding a spiral of cellulose nanocrystals for stimuli-responsive stretchable optics. *Nature Communications*, 10(1), 1–7.
- Lan, X., & Wang, Q. (2016). Self-assembly of chiral plasmonic nanostructures. *Advanced Materials*, 28(47), 10499–10507.
- Le, X., Lu, W., Zhang, J., & Chen, T. (2019). Recent progress in biomimetic anisotropic hydrogel actuators. *Advanced Science*, 6(5), 1801584.
- Lin, P., Zhang, T., Wang, X., Yu, B., & Zhou, F. (2016). Freezing molecular orientation under stretch for high mechanical strength but anisotropic hydrogels. *Small*, 12(32), 4386–4392.
- Liu, L., Li, L., Qing, Y., Yan, N., Wu, Y., Li, X., & Tian, C. (2016). Mechanically strong and thermosensitive hydrogels reinforced with cellulose nanofibrils. *Polymer Chemistry*, 7(46), 7142–7151.
- Mredha, M. T. I., Guo, Y. Z., Nonoyama, T., Nakajima, T., Kurokawa, T., & Gong, J. P. (2018). A facile method to fabricate anisotropic hydrogels with perfectly aligned hierarchical fibrous structures. *Advanced Materials*, 30(9), 1704937.
- Nyström, G., Arcari, M., & Mezzenga, R. (2018). Confinement-induced liquid crystalline transitions in amyloid fibril cholesteric tactoids. *Nature Nanotechnology*, 13(4), 330–336.
- Omidian, H., Hasherni, S.-A., Askari, F., & Nafisi, S. (1994). Swelling and crosslink density measurements for hydrogels. *Iranian Journal of Polymer Science and Technology*, 3(2).
- Park, N., & Kim, J. (2020). Hydrogel-based artificial muscles: Overview and recent progress. *Advanced Intelligent Systems*, 2(4), 1900135.
- Parker, R. M., Guidetti, G., Williams, C. A., Zhao, T., Narkevicius, A., Vignolini, S., & Frka-Petesic, B. (2018). The self-assembly of cellulose nanocrystals: Hierarchical design of visual appearance. *Advanced Materials*, 30(19), 1704477.
- Pasqui, D., De Cagna, M., & Barbucci, R. (2012). Polysaccharide-based hydrogels: The key role of water in affecting mechanical properties. *Polymers*, 4(3), 1517–1534.
- Pei, X., Zan, T., Li, H., Chen, Y., Shi, L., & Zhang, Z. (2015). Pure anisotropic hydrogel with an inherent chiral internal structure based on the chiral nematic liquid crystal phase of rodlike viruses. *ACS Macro Letters*, 4(11), 1215–1219.
- Qiao, L., Du, C., Gong, J. P., Wu, Z. L., & Zheng, Q. (2019). Programmed diffusion induces anisotropic superstructures in hydrogels with high mechano-optical sensitivity. *Advanced Materials Technologies*, 4(11), 1900665.
- Qijun, D., Jinsong, Z., Bin, W., Darong, T., & Kefu, C. (2018). Effect of nanocellulose fiber hornification on water fraction characteristics and hydroxyl accessibility during dehydration. *Carbohydrate Polymers*, 207, 44–51.
- Qin, H., Zhang, T., Li, N., Cong, H.-P., & Yu, S.-H. (2019). Anisotropic and self-healing hydrogels with multi-responsive actuating capability. *Nature Communications*, 10(1), 1–11.
- Saha, P., & Davis, V. A. (2018). Photonic properties and applications of cellulose nanocrystal films with planar anchoring. *ACS Applied Nano Materials*, 1(5), 2175–2183.
- Santiago-Gonzalez, B., Monguzzi, A., Capitani, C., Prato, M., Santambrogio, C., Meinardi, F., & Brovelli, S. (2018). Bottom-up synthesis and self-assembly of copper clusters into permanent excimer supramolecular nanostructures. *Angewandte Chemie International Edition*, 57(24), 7051–7055.
- Sun, Z., Yamauchi, Y., Araoka, F., Kim, Y. S., Bergueiro, J., Ishida, Y., & Aida, T. (2018). An anisotropic hydrogel actuator enabling earthworm-like directed peristaltic crawling. *Angewandte Chemie International Edition*, 57(48), 15772–15776.
- Tardy, B. L., Mattos, B. D., Otonari, C. G., Beaumont, M., & Rojas, O. J. (2021). Deconstruction and reassembly of renewable polymers and biocolloids into next generation structured materials. *Chemical Reviews*. <https://doi.org/10.1021/acs.chemrev.0c01333>
- Wang, P.-X., & MacLachlan, M. J. (2018). Liquid crystalline tactoids: Ordered structure, defective coalescence and evolution in confined geometries. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2112), 20170042.
- Whitesides, G. M., & Grzybowski, B. (2002). Self-assembly at all scales. *Science*, 295(5564), 2418–2421.
- Wu, Y., Jiang, Z., Zan, X., Lin, Y., & Wang, Q. (2017). Shear flow induced long-range ordering of rod-like viral nanoparticles within hydrogel. *Colloids and Surfaces B: Biointerfaces*, 158, 620–626.

- Yang, D.-K., Cui, Y., Nemati, H., Zhou, X., & Moheghi, A. (2013). Modeling aligning effect of polymer network in polymer stabilized nematic liquid crystals. *Journal of Applied Physics*, 114(24), Article 243515.
- Yang, Y., Tan, Y., Wang, X., An, W., Xu, S., Liao, W., & Wang, Y. (2018). Photothermal nanocomposite hydrogel actuator with electric-field-induced gradient and oriented structure. *ACS Applied Materials & Interfaces*, 10(9), 7688–7692.
- Ye, D., Cheng, Q., Zhang, Q., Wang, Y., Chang, C., Li, L., Peng, H., & Zhang, L. (2017). Deformation drives alignment of nanofibers in framework for inducing anisotropic cellulose hydrogels with high toughness. *ACS Applied Materials & Interfaces*, 9(49), 43154–43162.
- Yook, S., Es-Haghi, S. S., Yildirim, A., Mutlu, Z., & Cakmak, M. (2019). Anisotropic hydrogels formed by magnetically-oriented nanoclay suspensions for wound dressings. *Soft Matter*, 15(47), 9733–9741.
- Zeltmann, S. E., Prakash, K. A., Doddamani, M., & Gupta, N. (2017). Prediction of modulus at various strain rates from dynamic mechanical analysis data for polymer matrix composites. *Composites Part B: Engineering*, 120, 27–34.
- Zhou, L., Chen, M., Guan, Y., & Zhang, Y. (2014). Multiple responsive hydrogel films based on dynamic Schiff base linkages. *Polymer Chemistry*, 5(24), 7081–7089.
- Zhu, Q. L., Dai, C. F., Wagner, D., Daab, M., Hong, W., Breu, J., Zheng, Q., & Wu, Z. L. (2020). Distributed electric field induces orientations of nanosheets to prepare hydrogels with elaborate ordered structures and programmed deformations. *Advanced Materials*, 32(47), 2005567.